

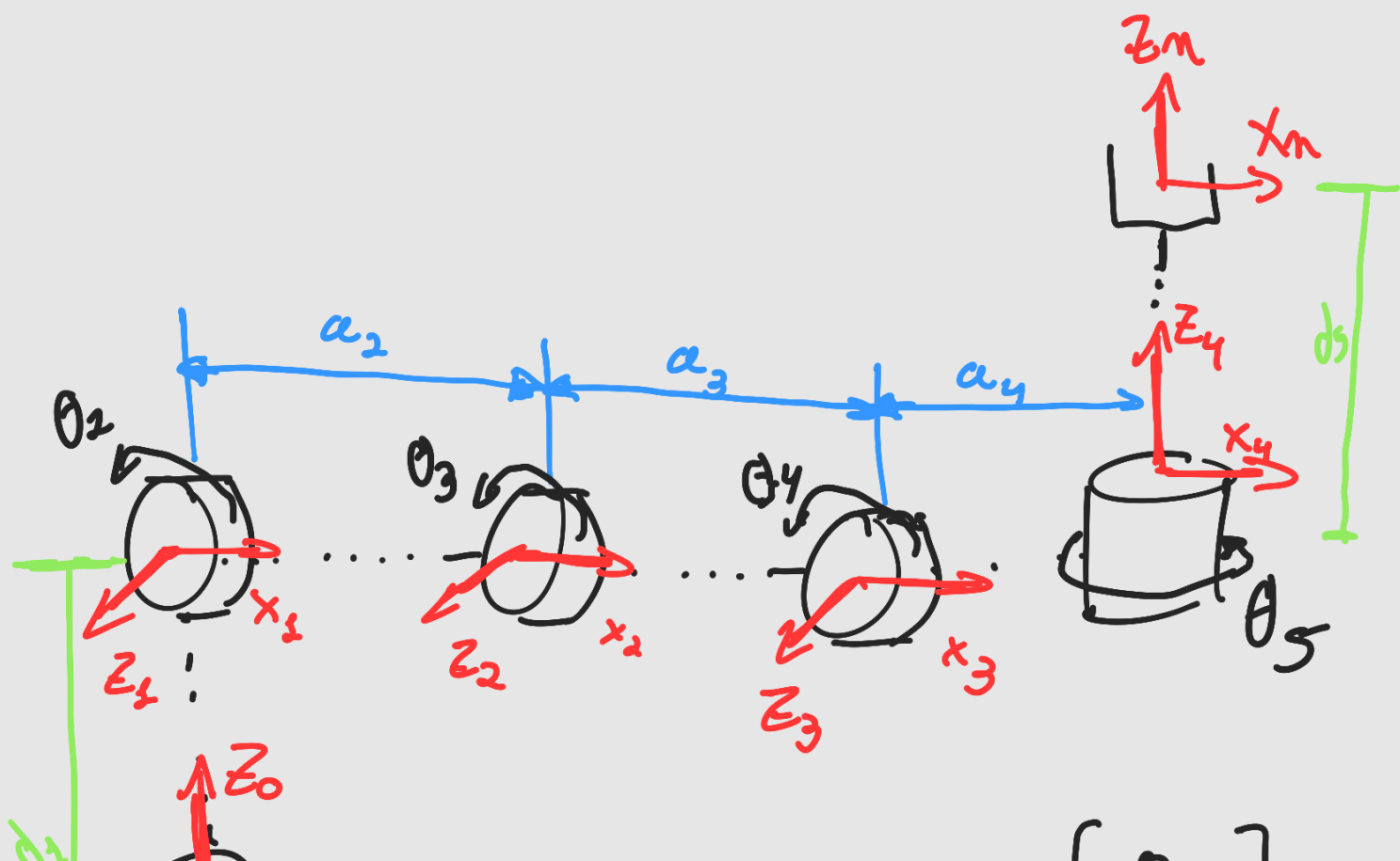
Fundamentos de Robótica

2023.2

Projeto - Deduções

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- Robô : 5 juntas Rotacionais





$$\vec{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \\ \theta_5 \end{bmatrix}$$

- Modelo Cinemático:

Tabela D.H

$J_{un/a}$	a_i	α_i	d_i	θ_i
1	0	$+90^\circ$	d_1	θ_1^*
2	a_2	0	0	θ_2^*
3	a_3	0	0	θ_3^*
4	a_4	-90°	0	θ_4^*
5	0	0	d_4	θ_5^*

* $\text{sen} = S$
 $\text{cos} = C$

Matriz D.H

$$A_i = A_i^{i-1} = \begin{bmatrix} C_{\theta_i} & -S_{\theta_i}C_{\alpha_i} & S_{\theta_i}S_{\alpha_i} & a_iC_{\theta_i} \\ S_{\theta_i} & C_{\theta_i}C_{\alpha_i} & -C_{\theta_i}S_{\alpha_i} & a_iS_{\theta_i} \\ 0 & S_{\alpha_i} & C_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_1(\theta_1) = \begin{bmatrix} C\theta_1 & 0 & S\theta_1 & 0 \\ S\theta_1 & 0 & -C\theta_1 & 0 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2(\theta_2) = \begin{bmatrix} C\theta_2 & -S\theta_2 & 0 & a_2 \cdot C\theta_2 \\ S\theta_2 & C\theta_2 & 0 & a_2 \cdot S\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3(\theta_3) = \begin{bmatrix} C\theta_3 & -S\theta_3 & 0 & a_3 \cdot C\theta_3 \\ S\theta_3 & C\theta_3 & 0 & a_3 \cdot S\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_4(\theta_4) = \begin{bmatrix} C\theta_4 & 0 & -S\theta_4 & a_4 \cdot C\theta_4 \\ S\theta_4 & 0 & C\theta_4 & a_4 \cdot S\theta_4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_5(\theta_5) = \begin{bmatrix} \cos & -\sin & 0 & 0 \\ \sin & \cos & 0 & 0 \\ 0 & 0 & 1 & d_5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_5^0 = A_1 \cdot A_2 \cdot A_3 \cdot A_4 \cdot A_5 \quad \left(\begin{array}{l} \text{calculated via} \\ \text{software} \end{array} \right)$$

